IOT-BASED SMART BATTERY MONITORING AND CHARGING SYSTEM

A PROJECT REPORT

Of

ECB1303 - EMBEDDED SYSTEM AND IOT DESIGN

Submitted by

MUTHUKUMAR B 8115U24EC064

NIROSHAN S 8115U24EC069

PRASANNA S 8115U24EC076

SAMUEL LOUIS J 8115U24EC094

in partial fulfilment for the award of the degree of

BACHELOR OF ENGINEERING

in

ELECTRONICS AND COMMUNICATION ENGINEERING

K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)
SAMAYAPURAM, TRICHY

ANNA
UNIVERSITY
CHENNAI 600 025

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Under the Guidance of

Dr.R.SAMSON DANIEL, M.Tech., Ph.D

Department of Electronics and Communication Engineering

K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)
Under
ANNA UNIVERSITY, CHENNAI

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BONAFIDE CERTIFICATE

Certified that this project report titled "**Iot-Based Smart Battery Monitoring And Charging System**" is the bonafide work of **MUTHU KUMAR B (8115U24EC064), NIROSHAN S (8115U24EC069) , PRASANNA S (8115U24EC076), SAMUEL LOUIS (8115U24EC094)** who carried out the project work under my supervision. Certified further, that to the best of my knowledge the work reported here in does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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DECLARATION BY THE CANDIDATES

We declare that to the best of our knowledge the work reported herein has been composed solely by ourselves and that it has not been in whole or in part in any previous application for a degree.

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DEPARTMENT OF ECE

VISION

To be distinguished as a prominent program in Electronics and Communication Engineering Studies by preparing students for Industrial Competitiveness and Societal Challenges.

MISSION

M1. To equip the students with latest technical, analytical and practical knowledge

M2. To provide vibrant academic environment and Innovative Research culture

M3. To provide opportunities for students to get Industrial Skills and Internships to meet out the challenges of the society.

PROGRAM EDUCATIONAL OBJECTIVES (PEO'S)

PEO1: Graduates will become experts in providing solution for the Engineering problems in Industries, Government and other organizations where they are employed.

PEO2: Graduates will provide innovative ideas and management skills to enhance the standards of the society by individual and with team works through the acquired Engineering knowledge.

PEO3: Graduates will be successful professionals through lifelong learning and contribute to the society technically and professionally.

PROGRAM SPECIFIC OUTCOMES (PSO'S):

PSO1: Analyze and design complex electronic circuits and communication systems by applying the principles of engineering and mathematics.

PSO2: Analyze and provide innovative design solutions to research problems, industrial problems and for the demanding needs of the society in the domains of electronic integrated circuits, signal processing, high frequency and antenna design.

PSO3: Continuously involve in life-long learning activities so as to evolve and become an active team member in diverse and challenging design situations.

PROGRAM OUTCOMES:

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Engineering tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and the World: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.

PO8: Individual and Collaborative Teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO9: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large. Some of them are, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO10: Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO11: Lifelong learning: Recognize the need for and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change.

CO-PO MAPPING FOR

EMBEDDED SYSTEMS AND IOT DESIGN

After successful completion of this course, the students should be able to

CO1: Analyze the internal architecture and modelling of an embedded processor using design tools.

CO2: Develop the embedded system networks using different protocols.

CO3: Identify the evolution of the Internet of Things (IoT) using internal components for networking.

CO4: Evaluate cloud and big data analytics using IoT Platform.

	POs											PSOs			
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3	Average
CO1	2.60	1.40	2.00	1.80	1.80	3.00	1.40	1.80	1.40	2.00	3.00	1.80	1.40	0.80	1.87
CO2	2.60	1.20	2.00	1.20	2.00	3.00	1.20	1.40	1.60	1.80	3.00	1.80	1.80	0.20	1.77
CO3	2.60	1.20	2.00	1.60	1.00	3.00	1.20	1.60	1.80	1.20	3.00	1.80	1.80	0.80	1.76
CO4	2.60	1.20	1.20	2.00	1.20	3.00	1.20	1.80	1.60	1.20	3.00	1.80	1.80	1.20	1.77
CO5	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00
Average	2.68	1.60	2.04	1.92	1.80	3.00	1.60	1.92	1.88	1.84	3.00	2.04	1.96	1.20	2.03

SUSTAINABLE DEVELOPMENT GOALS (SDG) MAPPING

The proposed project,“ **Iot-Based Smart Battery Monitoring And Charging System**” contributes to the United Nations Sustainable Development Goals (SDGs) through intelligent battery monitoring, energy management, and electronic protection.

SELECTED SUSTAINABLE DEVELOPMENT GOALS

SDG 7 – Affordable and Clean Energy: The project supports efficient battery monitoring and energy management by measuring electrical parameters such as voltage, current, and power. The MOSFET-based switching mechanism helps control the battery load during programmed protection conditions. This approach can support reliable battery backup and energy storage applications.

SDG 9 – Industry, Innovation and Infrastructure: The project applies ESP32-based embedded technology, sensor integration, and power MOSFET switching to develop an intelligent battery monitoring and protection prototype. It demonstrates an innovative approach that can be further developed for industrial backup power systems and other energy storage applications.

ACKNOWLEDGEMENT

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We thank **Dr. M. MAHESWARI, M.Tech., Ph.D.,** Head of the Department of **ELECTRONICS AND COMMUNICATION ENGINEERING**, for providing her encouragement in pursuing this project.

We wish to convey our profound and heartfelt gratitude to our esteemed project guide and Course Coordinator **Dr.R.SAMSON DANIEL, M.Tech., Ph.D,** Assistant Professor, Department of **ELECTRONICS AND COMMUNICATION ENGINEERING,** for her incalculable suggestions, creativity, assistance and patience, which motivated us to carry out this project.

We wish to express our special thanks to the officials and Lab Technicians of our departments who rendered their help during the period of the work progress.

ABSTRACT

Battery backup systems are widely used in Uninterruptible Power Supply (UPS) systems, industrial equipment, and telecommunication networks. This project proposes Iot-Based Smart Battery Monitoring And Charging System for Industrial Applications Using ESP32. The system is designed to monitor the voltage, charging and discharging current, and temperature of a 12 V lead-acid battery using a voltage divider, INA219 current sensor, and DS18B20 temperature sensor. An ESP32 microcontroller processes the sensor data and displays the measurements on an OLED display. Power MOSFETs are used for electronic switching to control the charging and load paths. A suitable Constant Current/Constant Voltage (CC/CV) charging circuit is incorporated according to the battery manufacturer's specifications. The proposed system provides protection against low battery voltage, excessive charging voltage, overcurrent, and overtemperature conditions. LEDs and a buzzer indicate the operating status and faults. The system is also designed to supply the connected load from the battery during DC adapter failure, provided that safe operating conditions are maintained. This project aims to improve battery monitoring, protection, and operational reliability through embedded control and power electronics. The design will be evaluated through circuit simulation and hardware testing, where implemented, to verify its performance.

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CHAPTER-1
INTRODUCTION

INTRODUCTION

Battery systems are widely used in industrial equipment, emergency power units, communication systems, automation panels, renewable energy systems, and portable electrical devices. A battery stores electrical energy and supplies power when the main supply is unavailable or when temporary power is required. Although batteries are useful energy sources, they must be monitored and protected during charging and discharging. Incorrect charging, excessive current, deep discharge, and high temperature can reduce battery life and may create unsafe operating conditions.

A battery management and protection system is used to observe the condition of a battery and control its operation. The main purpose of such a system is to keep the battery within a safe voltage, current, and temperature range. If an abnormal condition occurs, the system should provide a warning and disconnect the load when necessary. A good protection system should also provide clear information to the user so that the reason for the warning or shutdown can be understood.

The project titled “Power MOSFET-Based Intelligent Battery Management and Protection System for Industrial Applications Using ESP32” focuses on the development of an intelligent monitoring and protection concept. The ESP32 microcontroller is used as the main control unit. It receives battery-related values such as voltage, current, and temperature, compares these values with selected safety limits, and decides whether the battery load should remain connected or be disconnected.

A power MOSFET is used as the electronic switching device. The MOSFET is controlled by a signal from the ESP32. During normal operation, the MOSFET allows the load to remain connected to the battery. When an unsafe condition is detected, the ESP32 changes the MOSFET control signal and disconnects the load. This provides faster and more controlled switching compared with a simple mechanical switch.

The proposed system also includes an OLED display, LEDs, and a buzzer. The OLED display presents the measured values and the operating status of the battery system. The LEDs provide a quick visual indication of normal, warning, and fault conditions. The buzzer provides an audible alert when immediate attention is required. By combining these devices, the system becomes easier to monitor and operate.

The project output is demonstrated using Wokwi simulation. The project does not claim that a complete physical prototype was constructed or tested. Wokwi is used to demonstrate the ESP32 program, simulated voltage, current, and temperature values, OLED display operation, LED indications, buzzer alerts, and MOSFET control logic. The simulation helps verify the relationship between the input conditions and the programmed protection response.

The simulation allows different battery conditions to be applied without using a real battery or industrial load. Normal operation, low voltage, high current, and high temperature conditions can be represented and observed. The response of the controller can then be checked through the OLED display, LEDs, buzzer, and MOSFET control output.

However, the Wokwi simulation does not prove the actual electrical, thermal, mechanical, or safety performance of a physical battery protection system. It does not represent the complete behaviour of a real battery, actual sensor accuracy, MOSFET

heating, wiring resistance, connector failure, electrical arcs, electromagnetic interference, or short-circuit energy. These factors would require separate physical design and testing.

The main purpose of the proposed project is to demonstrate how an embedded controller can combine monitoring, decision-making, indication, and electronic switching. The system provides a clear foundation for future hardware development while maintaining a clear difference between simulation results and physical test results.

1.1 EXISTING SYSTEM

1.1.1 Traditional Battery Monitoring and Protection System

Traditional battery monitoring and protection systems use different electrical devices to maintain safe battery operation. Common devices include fuses, circuit breakers, relays, contactors, voltage meters, current sensors, and temperature protection circuits. The selection of these devices depends on battery voltage, current rating, battery chemistry, and application.

In basic systems, a fuse and manual switch are used to protect and control the battery. Advanced systems may include separate monitoring and protection circuits to measure battery parameters and disconnect the load during abnormal conditions. These systems are commonly used in UPS units, industrial equipment, vehicles, and backup power applications.

1.1.2 Fuse, Circuit Breaker, Relay and Contactor-Based Protection

A fuse is one of the simplest protection devices used in battery systems. It contains a metal element that melts when excessive current flows through it, thereby disconnecting the circuit. A circuit breaker performs a similar function but can normally be reset after the fault is cleared.

Relays and contactors are used to control the connection between the battery and the load. They operate using electromagnetic coils and mechanical contacts. Relays are commonly used in low- and medium-power switching applications, while contactors are designed for higher-current applications.

Although these devices provide basic electrical protection, they have some limitations. Mechanical contacts may wear out after repeated switching, and electrical arcing may occur when disconnecting high-current loads. Relay coils also consume power, and switching speed is limited by mechanical movement.

1.1.3 Dedicated Battery Monitoring and Protection Circuits

Dedicated battery protection circuits are designed to monitor important battery parameters such as voltage, current, and temperature. These circuits help identify abnormal conditions, including overcharging, over-discharging, excessive current, and short circuits. Some protection circuits use MOSFETs to automatically disconnect the battery when unsafe conditions are detected.

Voltage monitoring helps identify low or high battery voltage conditions. Current monitoring is used to detect overload and excessive charging or discharging current. Temperature monitoring helps identify overheating that may affect battery performance and safety.

These circuits are compact, reliable, and widely used in rechargeable battery packs. However, their internal protection logic may be difficult to modify. Adding customized warning messages, displays, and additional monitoring functions may require extra components and circuits.

1.1.4 Limitations of Existing Battery Systems

Traditional battery monitoring systems often use separate circuits for voltage measurement, current monitoring, temperature protection, and load switching. As a

result, the operator may need to check different devices to understand the complete battery condition.

Manual inspection is also required in some systems, which may cause delays in identifying abnormal conditions. Basic protection devices such as fuses and circuit breakers cannot continuously display battery operating parameters. Mechanical switching devices may also experience contact wear and switching delays.

Some dedicated protection circuits provide limited flexibility for adding customized monitoring and display functions. Therefore, an integrated system using a microcontroller, sensors, and Power MOSFET can provide better monitoring, faster control, and clear indication of battery conditions.

1.2 DRAWBACKS OF EXISTING SYSTEM

1.2.1 Limited Continuous Monitoring and Delayed Fault Detection

One of the major drawbacks of conventional battery protection systems is the lack of continuous monitoring. Fuses and circuit breakers can disconnect the circuit during excessive current conditions, but they cannot continuously display battery voltage, current, or temperature. Manual monitoring also depends on the availability of an operator.

In industrial applications, batteries may operate for several hours without direct supervision. During this period, voltage may decrease, current may increase, or temperature may rise. If these conditions are not monitored continuously, faults may be detected only after affecting the battery or connected equipment. Therefore, continuous monitoring using a microcontroller can help identify abnormal conditions at an early stage.

1.2.2 Limitations of Mechanical Switching Devices

Traditional systems commonly use relays and contactors for controlling battery connections. These devices operate through electromagnetic coils and mechanical

contacts. However, repeated switching can cause contact wear and mechanical damage. Contact bounce and electrical arcing may also occur during switching, especially when disconnecting high-current loads.

The relay coil consumes additional power when energized, and its switching speed is limited by mechanical movement. In comparison, a Power MOSFET performs electronic switching without mechanical contacts. It can provide faster switching and support repeated operation when properly selected and operated within its electrical and thermal ratings.

1.2.3 Limited Fault Identification and Monitoring Flexibility

Conventional protection devices generally perform individual functions. A fuse mainly protects against excessive current, a thermal switch responds to temperature, and a voltage indicator displays battery voltage. These separate devices may not combine all the measured parameters to identify the complete battery condition.

Another drawback is limited fault information. When a fuse opens or a relay disconnects the battery, the operator may not know the exact reason for the protection action. Conventional systems may also have limited flexibility for modifying warning levels, protection conditions, and display functions.

A microcontroller-based system provides greater flexibility because its control logic can be modified through software. An OLED display can show battery readings and fault messages, while LEDs and a buzzer can provide visual and audible indications.

1.2.4 Safety, Maintenance and Simulation Limitations

Simple battery protection arrangements may create safety risks if components are incorrectly selected or connected. Incorrect wiring, poor connections, excessive current, and inadequate heat dissipation can affect system performance. MOSFETs also require proper selection based on voltage rating, current rating, gate voltage, on-

resistance, and thermal performance.

Manual inspection and mechanical protection devices may increase maintenance requirements. Identifying multiple faults at the same time can also be difficult when separate protection circuits operate independently.

Simulation tools such as Wokwi can demonstrate the basic software logic and system operation. However, simulation cannot fully represent actual battery behaviour, high fault currents, component heating, electrical arcing, or all physical safety conditions. Therefore, practical hardware testing and suitable battery protection circuits are necessary for real-world applications.

1.3 NEED FOR THE PROPOSED WORK

1.3.1 Need for Intelligent Battery Monitoring

Modern battery systems require more than simple ON/OFF control. Continuous monitoring of voltage, current, and temperature is important for maintaining safe and reliable battery operation. Traditional protection devices mainly disconnect the circuit during abnormal conditions but may not provide complete information about battery status.

The proposed system combines monitoring and protection functions using a programmable controller. It helps identify abnormal operating conditions, provides information to the user, and controls the connected load according to programmed conditions. This approach can reduce dependence on manual inspection and separate monitoring devices.

1.3.2 Need for ESP32-Based Monitoring and Control

The ESP32 is used as the main controller because it has multiple input and output pins and supports programmable control operations. It receives sensor information, processes the measured values, updates the OLED display, controls LED indicators, and generates switching signals for the Power MOSFET.

The ESP32 also provides flexibility for future development. Additional features such as wireless communication, data logging, remote monitoring, and real-time clock functionality can be integrated. This makes it a suitable platform for developing an intelligent battery monitoring system.

1.3.3 Need for Power MOSFET-Based Protection

A Power MOSFET is required because the ESP32 cannot directly handle the high current required by a battery-connected load. The ESP32 generates a low-power control signal, while the MOSFET performs electronic switching in the power circuit.

During normal operation, the MOSFET allows the load to remain connected. When an abnormal condition is detected, the ESP32 changes the MOSFET control signal to disconnect the load. MOSFET switching is suitable for repeated operation because it has no mechanical contacts and can operate at high switching speeds when correctly selected.

The MOSFET must be selected according to the battery voltage, load current, gate voltage, on-resistance, and heat dissipation requirements. Proper electrical and thermal protection is also necessary in a practical system.

1.3.4 Need for User Indication, Fault Detection and Testing

An effective battery monitoring system should provide clear information about its operating condition. An OLED display can show voltage, current, temperature, and system status. LED indicators provide quick visual information, while a buzzer can provide an audible warning when a fault occurs.

The proposed control system can be organized into three operating states:

- Normal state: All measured values remain within the selected operating limits, and the load remains connected.
- Warning state: A parameter approaches its protection limit, and the system provides a warning indication.
- Fault state: A parameter crosses its protection limit, and the controller disconnects the load through the MOSFET.

Simulation allows different operating conditions to be tested, including low voltage, excessive current, high temperature, and recovery conditions. It helps verify the programmed control logic before hardware implementation. However, actual battery safety, sensor accuracy, MOSFET heating, and electrical performance must be verified through physical testing.

1.4 APPLICATIONS OF PROPOSED SYSTEM

The proposed system can be used in industrial battery backup systems to monitor battery voltage, current, and temperature. It can disconnect the load when the battery voltage becomes too low or when an unsafe condition is detected.

It can be used in industrial control panels and automation equipment. The ESP32 can display the battery condition on the OLED, while the LEDs and buzzer alert the operator about warning or fault conditions.

The system is also suitable for communication equipment and remote monitoring units. It can help protect batteries used in communication towers, remote sensors, and emergency communication systems from over-discharge and overcurrent.

The proposed system can be applied to renewable energy storage systems such as small solar power systems. It can monitor the battery during charging and discharging and control the connected load using the power MOSFET.

It may also be used in emergency lighting systems and auxiliary battery systems of electric vehicles. The system can provide automatic protection and clear indication when the battery voltage, current, or temperature becomes abnormal.

In this project, these applications are demonstrated only through Wokwi simulation.

The simulation shows the ESP32 control logic, MOSFET switching, OLED display, LED indications, and buzzer alerts. A physical industrial application would require further hardware design, calibration, safety testing, and validation.

CHAPTER-2

LITERATURE SURVEY

2.1. LITERATURE SURVEY

1. **Advances in Batteries, Battery Modeling, Battery Management System, Battery Thermal Management, SOC, SOH, and Charge/Discharge Characteristics in EV Applications — R. Ranjith Kumar et al. — 2023**

Key Findings:

- Reviews battery management systems, battery modelling, charging methods, and thermal management.
- Explains state-of-charge (SoC) and state-of-health (SoH) estimation techniques.
- Discusses voltage and current monitoring, battery protection, and charging/discharging control.
- Identifies research challenges in improving battery safety and performance.

Disadvantages:

- The study is primarily a review and does not provide a single ready-to-build circuit for the proposed project.
- Some advanced battery models require significant computational resources.
- Additional hardware implementation is required to validate a practical 12 V lead-acid battery protection system.

2. **Battery Management System: An Overview of Its Application in the Smart Grid and Electric Vehicles — H. Rahimi-Eichi, U. Ojha, F. Baronti and M. Y. Chow — 2013**

Key Findings:

- Explains the importance of battery management in smart grids and electric vehicles.
- Highlights deep-discharge protection and accurate SoC and SoH estimation.

- Discusses battery monitoring algorithms and safety mechanisms.
- Establishes the need for reliable battery management to improve battery performance and service life.

Disadvantages:

- The paper provides a broad overview rather than a detailed implementation of a low-cost embedded protection circuit.
- Advanced estimation methods may require calibration and battery-specific models.
- Additional design work is needed to implement automatic MOSFET-based load disconnection..

3. **IoT-Enabled Monitoring System for Battery Management in Inverter Systems — Blessing Nneka Ben-Festus and Olugbenga A. Falode — 2026**

   Key Findings:

   - Develops an IoT-enabled battery monitoring system for inverter applications.
   - Monitors battery voltage, current, and temperature using sensors and a microcontroller.
   - Uses a GSM module and cloud platform for remote monitoring.
   - Reports experimental measurements and evaluates monitoring accuracy and communication performance.

   Disadvantages:

   - The reported implementation uses an Arduino-based controller and GSM communication rather than the proposed ESP32-based MOSFET switching architecture.
   - Cloud monitoring depends on communication availability.
   - A separate assessment is required to establish the suitability of the design for industrial loads and the selected lead-acid battery.

4. **Challenges and Strategies for Implementation of Lead-Acid Battery Health Monitoring and Prediction in Off-Grid Solar Panel Systems — 2025**

Key Findings:

- Investigates lead-acid battery health monitoring for off-grid solar applications.
- Uses an equivalent circuit model and linear regression to estimate internal resistance and open-circuit voltage.
- Examines the relationship between battery degradation and internal resistance.
- Discusses practical monitoring challenges, including data loss caused by network interruptions.

Disadvantages:

- The focus is on battery health estimation rather than complete real-time power-path protection.
- Estimation accuracy depends on the measurements and assumptions used in the battery model.
- Direct integration with MOSFET-based overcurrent and low-voltage disconnection requires further development.

5. **IoT-Based Monitoring of Sealed Lead-Acid Battery Charging to Prevent Overheating and Overcharging — Rizal W. H. Siagian et al. — 2025**

Key Findings:

- Uses an ESP32 to monitor sealed lead-acid battery charging.
- Uses the INA219 sensor for voltage and current measurements and the DS18B20 for temperature monitoring.
- Sends measurement data to Firebase and displays it through an Android application.
- Evaluates measurement errors for 12 V and 6 V batteries.

Disadvantages:

- The main focus is charging monitoring and mobile visualization.
- The reported measurement errors indicate that calibration and sensor placement need attention.
- A complete industrial backup system with independently verified MOSFET load isolation and adapter-failure handling requires additional design work.

6. **A Review of Battery Energy Storage Systems and Advanced Battery Management**

System for Different Applications: Challenges and Recommendations — Shaik Nyamathulla and C. Dhanamjayulu — 2024

Key Findings:

- Reviews battery energy storage technologies and their applications.
- Discusses SoC, SoH, and remaining useful life estimation.
- Examines advanced battery management functions and their technical challenges.
- Provides recommendations for improving battery reliability and energy storage performance.

Disadvantages:

- The paper covers multiple battery technologies and does not focus exclusively on 12 V lead-acid batteries.
- Some advanced management techniques may be too complex for a low-cost student prototype.
- A practical implementation still needs component selection, protection thresholds, and experimental validation.

7. **Advances in Battery State Estimation of Battery Management System in Electric Vehicles — 2024**

Key Findings:

- Reviews methods for estimating battery SoC and SoH.
- Examines battery degradation, state-estimation models, and different BMS architectures.
- Discusses onboard, cloud-based, and related battery management approaches.
- Identifies future research opportunities in battery state estimation.

Disadvantages:

- The review mainly addresses lithium-ion batteries and electric vehicle applications.
- Advanced estimation methods may require extensive data and calibration.
- The proposed project needs a simpler monitoring and threshold-based protection algorithm suitable for a 12 V lead-acid battery.

8. **Applications of Artificial Intelligence and Cell Balancing Techniques for**

Battery Management System in Electric Vehicles: A Comprehensive Review — Arunesh Kumar Singh et al. — 2024

Key Findings:

- Reviews artificial intelligence techniques for battery management.
- Discusses fuzzy logic, support vector machines, neural networks, and optimization methods.
- Explains cell balancing and its role in battery performance and safety.
- Identifies opportunities for improving SoC estimation and battery operation.

Disadvantages:

- AI-based methods can require large datasets and additional processing resources.
- Cell balancing is more relevant to multi-cell battery packs than to a single 12 V lead-acid battery.
- The proposed project prioritizes direct sensor measurement and protective switching instead of computationally intensive AI algorithms.

9. **Battery Health Management—A Perspective of Design, Optimization, Manufacturing, Fault Detection, and Recycling — Pavel M. Roy et al. — 2024**

Key Findings:

- Reviews battery design, optimization, and health management.
- Covers lead-acid, lithium-ion, and nickel-metal hydride batteries.
- Discusses fault detection and the use of AI and machine learning for battery diagnostics.
- Highlights the importance of early fault identification and battery lifecycle management.

Disadvantages:

- The review covers a broad range of battery technologies and lifecycle topics.
- Advanced fault-detection methods may require additional sensors and computational resources.
- Direct implementation of predictive fault detection is outside the basic scope of the proposed prototype.

10. **A Battery Management System with EIS Monitoring of Life Expectancy for**

Lead–Acid Batteries — 2021

Key Findings:

- Presents a battery management system for lead-acid batteries.
- Uses electrical impedance spectroscopy (EIS) to assess battery condition.
- Integrates voltage, temperature, and impedance measurements.
- Explores the use of impedance information for estimating battery life expectancy.

Disadvantages:

- EIS requires additional measurement hardware and signal-processing techniques.
- The implementation is more complex than basic voltage, current, and temperature monitoring.
- The proposed project uses a simpler sensing arrangement to keep the system affordable and practical for a student prototype.

11. **Concept of a Cloud State Modeling System for Lead-Acid Batteries: Theory and Prototyping — 2021**

Key Findings:

- Examines cloud-based modelling of lead-acid battery condition.
- Discusses battery ageing and the difficulty of identifying deeply discharged or ageing batteries.
- Presents an approach for monitoring battery condition and exploring future failure prediction.
- Highlights the potential of remote monitoring for reducing unexpected battery failures.

Disadvantages:

- The work emphasizes monitoring and prediction rather than complete local hardware protection.
- Cloud-based analysis depends on communication and data availability.
- The proposed project needs local protection that continues operating even when the network is unavailable.

12. **Design and Implementation of a Smart Battery Management System Using ESP-Based IoT and Sensor Integration — 2025/2026**

Key Findings:

- Uses an ESP-family microcontroller and sensors to monitor battery voltage and temperature.
- Describes real-time data acquisition and programmed monitoring.
- Discusses the use of sensor information to identify potentially unsafe battery conditions.
- Provides a relevant reference for implementing an ESP32-based monitoring architecture.

Disadvantages:

- The available publication record provides limited detail for independently assessing all implementation results.
- Its reported scope does not establish that the exact proposed lead-acid battery, CC/CV charger, and MOSFET switching circuit has been validated.
- Additional circuit-level design and testing are needed to demonstrate reliable industrial backup operation.

PROJECT DESCRIPTION

3.1 PROPOSED SYSTEM DESCRIPTION

The proposed project, “Power MOSFET-Based Intelligent Battery Management and Protection System for Industrial Applications Using ESP32,” is developed to monitor, control, and protect a 12 V lead-acid battery used in industrial backup power applications. Batteries are important components in uninterrupted power systems because they provide electrical energy when the main power supply becomes unavailable. However, improper charging, excessive discharge, high current, and abnormal temperature can affect battery performance and reduce its service life. Therefore, an intelligent monitoring and protection system is required to maintain reliable battery operation and protect the connected electrical equipment.

The proposed system uses an ESP32 microcontroller as the central processing and control unit. It collects information from different sensors and evaluates the operating condition of the battery. The system continuously monitors battery voltage, current, and temperature to identify abnormal conditions. The controller processes these measurements and compares them with predefined operating limits. Based on the measured values, the system determines whether the battery is operating normally or whether protective action is required. This approach reduces dependence on manual inspection and provides a convenient method for monitoring the battery condition.

The system consists of a 12 V lead-acid battery, a DC adapter, a suitable constant-current/constant-voltage charging circuit, power MOSFETs, an INA219 current sensor, a DS18B20 temperature sensor, a voltage-sensing circuit, an OLED display, status LEDs, and a buzzer. Each component performs a specific function in the monitoring and protection process. The DC adapter provides external power, while the charging circuit regulates the charging process according to the battery specifications. The battery stores electrical energy and supplies the connected load when external power is unavailable.

The battery-voltage sensing circuit measures the terminal voltage and provides a suitable signal to the ESP32. The INA219 sensor measures current and bus voltage at its installed measurement point, allowing the controller to identify abnormal current consumption. The DS18B20 temperature sensor provides digital temperature measurements that help identify excessive heating. These measurements are processed by the controller to determine the operating status of the battery. The OLED display presents the measured values, while the LEDs and buzzer provide visual and audible indications of normal operation and fault conditions.

Power MOSFETs are used as electronic switching devices to control the selected charging or load path. They provide fast switching and can reduce conduction losses when correctly selected and operated. When the controller identifies an unsafe condition, it sends a control signal through a suitable gate-drive circuit to change the switching state. For example, if the battery voltage falls below the configured low-voltage limit, the system can disconnect the load to reduce the risk of excessive discharge. Similarly, overcurrent and overtemperature conditions can trigger protective action according to the programmed control logic.

The proposed system is intended for small industrial backup applications, including UPS units, telecommunication equipment, industrial control panels, emergency lighting, and low-voltage energy storage systems. It combines embedded programming, sensor interfacing, power electronics, and fault detection in a single prototype. The main aim is to improve monitoring, provide timely fault alerts, and demonstrate automatic battery protection. The effectiveness of the system depends on correct circuit design, appropriate component selection, accurate sensor calibration, and testing under different operating conditions.

3.2 CIRCUIT DIAGRAM AND ITS EXPLANATION

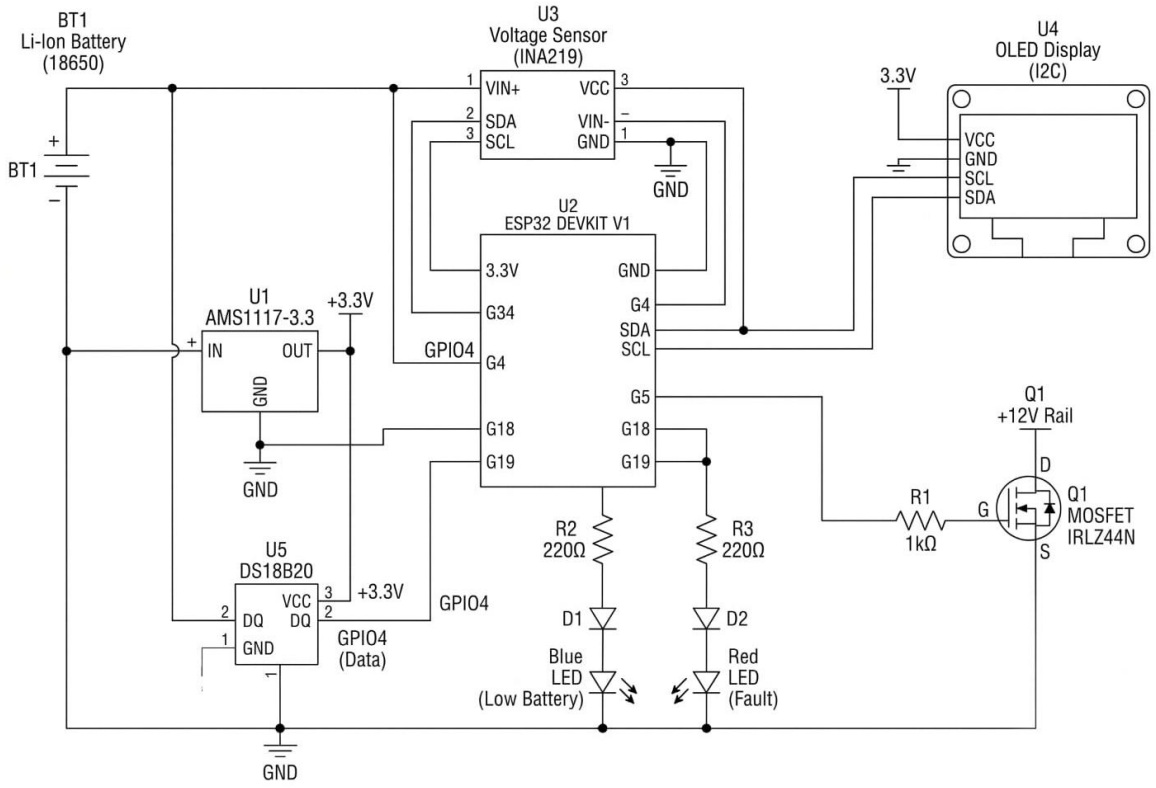


Fig 3.1 Circuit Diagram

The circuit diagram represents the electrical connections between the battery, external power supply, charging circuit, microcontroller, sensors, switching devices, and output indicators. The circuit is divided into several functional sections: the input power section, battery charging section, sensing section, control section, MOSFET switching section, and indication section. Each section performs a specific task, and all sections work together to monitor the battery and respond to abnormal conditions.

The input power section consists of a DC adapter that provides the required external DC supply. The adapter voltage must match the input requirements of the selected charging module and power-conversion circuit. A suitable regulated supply is used to power the ESP32 and its associated low-voltage components. The battery is connected through an appropriately rated power path so that it can supply the load during an external supply failure. Protective components are required to prevent damage

caused by reverse polarity, excessive current, or incorrect connections.

The battery charging section consists of a suitable CC/CV charger designed for the selected 12 V lithium ion battery. During charging, the circuit regulates the current and voltage according to the battery manufacturer's recommendations. The charging profile must be appropriate for the battery type, capacity, and operating temperature. A generic CC/CV module should not automatically be assumed suitable for every lead-acid battery, because charging voltage and termination requirements can differ between flooded, AGM, and gel batteries. The charger should also include appropriate safeguards against overcharging and excessive temperature.

The voltage-sensing section measures the battery terminal voltage. Since the battery voltage may exceed the ESP32's analog-input range, a resistor voltage divider is used to scale the voltage to a safe level. The scaled voltage is connected to an ESP32 analog input, where the controller obtains a digital measurement using its analog-to-digital converter. The software uses the divider ratio to estimate the original battery voltage. Correct resistor selection, calibration, grounding, and input protection are important for obtaining reliable measurements.

The current-sensing section uses the INA219 sensor to measure current and bus voltage at the selected point in the circuit. The sensor communicates with the ESP32 through the I^2C communication interface. Its measurements help the controller identify whether the load current is within the permitted operating range. The INA219 must be connected according to its shunt-current rating, bus-voltage rating, and intended current direction. It should not be treated as the only protection device against a sudden short circuit, because its measurement and software response may be slower than the fault current rise.

The temperature-sensing section uses the DS18B20 digital temperature sensor. It measures temperature at the selected monitoring point and transfers the result to the

ESP32 through a one-wire digital interface. A suitable pull-up resistor is required on the data line. The sensor may be positioned near the battery or a selected component to monitor local temperature, but the measured temperature does not necessarily represent the internal temperature of the battery. Sensor placement must therefore be considered when defining temperature limits and interpreting readings.

The control section consists of the ESP32 microcontroller and the embedded program. The controller receives sensor readings, evaluates operating conditions, and determines the required switching and alert states. If the measurements remain within the configured limits, the system continues normal monitoring. If a fault is identified, the controller activates the relevant LED and buzzer and commands the MOSFET switching circuit to disconnect the load or disable the selected path. Software filtering, fault confirmation, and recovery conditions can help prevent unnecessary switching caused by measurement noise.

The MOSFET switching section controls the intended electrical path. A suitable MOSFET is selected based on its voltage rating, current capacity, on-state resistance, gate-drive requirements, and thermal performance. The ESP32 produces a control signal, but the MOSFET gate must be driven correctly relative to its source terminal. Depending on the circuit topology, a dedicated gate driver may be required. Gate resistors, gate pull-down resistors, appropriate heat dissipation, and protection against switching transients are also important. For bidirectional blocking or isolation requirements, a single MOSFET may be insufficient because its body diode can permit current in one direction.

The indication section contains the OLED display, LEDs, and buzzer. The OLED provides information such as battery voltage, current, temperature, adapter status, and fault messages. The LEDs indicate operating states, such as normal operation, charging, or fault detection. The buzzer provides an audible warning when an abnormal condition is detected. These indicators help the user understand the condition of the system

without requiring continuous observation of the circuit.

The overall circuit works through coordination between the sensing, control, and switching sections. The sensors collect battery information, the ESP32 evaluates the measurements, and the MOSFET driver operates the switching devices according to the programmed logic. The charging circuit regulates charging independently, while the controller supervises the operating condition and reports faults.

3.3 COMPONENTS USED

Table 3.1 Components Used

S. No.	Component	Function
1	ESP32 DevKit V1	Main controller
2	Lithium-ion battery	Main power source
3	INA219 current sensor	Current, voltage and power monitoring
4	OLED display (0.96-inch)	Displays battery readings and system operating status
5	Power MOSFET (IRLZ44N)	Controls the load through electronic switching
6	Voltage regulator (AMS1117-5.0)	Regulates the input voltage to a stable 5V output
7	Potentiometer	Provides adjustable voltage input for demonstration and testing
8	Blue LED	Low-voltage status indication
9	Red LED	Fault status indication
10	Breadboard	Provides a platform for mounting components and establishing electrical connections

3.4 SOFTWARE USED

3.4.1 Arduino IDE

The Arduino IDE is used to develop and upload the control program to the ESP32 development board. It provides a programming environment for writing, compiling, and debugging embedded C/C++ code. It also supports libraries that simplify communication with sensors and display modules.

The program begins by initializing the ESP32 GPIO pins, communication interfaces, sensors, display, and warning devices. It then repeatedly reads the battery voltage, current, and temperature. The measurements are processed and compared with the protection thresholds. According to the results, the software updates the display, controls the LEDs and buzzer, and sends commands to the MOSFET driver.

The program should include suitable error handling for sensor communication failures and invalid readings. It should also define safe output states during startup, reset, and recovery. These measures are important because an incorrect sensor value or unexpected software state could otherwise lead to an inappropriate switching decision.

3.4.2 Embedded C/C++ Programming

Embedded C/C++ is used to implement the control logic of the battery management system. The program contains instructions for sensor data acquisition, voltage calculation, current monitoring, temperature checking, adapter-status detection, and output control.

The measured values are compared with the configured thresholds. When all parameters remain within the permitted operating range, the system continues normal monitoring. When an abnormal condition is detected, the software identifies the fault, activates the relevant warning, and requests the appropriate switching action.

The program should use clear fault-handling states rather than relying only on individual conditional statements. For example, a low-voltage fault can remain active until the battery voltage rises above a separate recovery threshold. This prevents repeated switching when the voltage fluctuates near the protection limit. Similar logic can be applied to temperature and other recoverable faults.

3.4.3 Sensor Libraries

Suitable software libraries are used to communicate with the INA219 sensor, DS18B20 temperature sensor, and OLED display. These libraries provide functions for reading measurements, converting sensor data, and displaying information. Using established libraries can reduce programming complexity and improve code readability.

3.4.4 Wokwi Simulator

Wokwi is an online simulation platform used to design and test the ESP32-based battery management system. It allows the ESP32 program to be executed with supported virtual components. In this project, Wokwi is used to demonstrate battery monitoring logic, display updates, fault detection, LED indications, and buzzer alerts. Different input values can be tested to observe the system's response under normal and fault conditions. The simulation helps verify the program's control logic without requiring the complete physical hardware setup.

3.5 ADVANTAGES OF THE PROPOSED SYSTEM

Real-Time Monitoring

The system continuously monitors battery voltage, current, and temperature. This information helps the user understand the battery's operating condition during charging, discharging, and backup operation. Real-time measurements also provide a useful basis for identifying abnormal conditions before they cause further problems.

Automatic Fault Detection

The ESP32 compares the sensor measurements with predefined protection thresholds. It can identify low-voltage, overvoltage, overcurrent, and overtemperature conditions and activate the corresponding warnings. Automatic fault detection reduces dependence on manual observation and makes the prototype more convenient to operate.

Electronic Load Control

Power MOSFETs provide electronic switching for the selected load or charging path. When correctly selected and driven, they can provide rapid switching and low conduction losses. The switching devices allow the controller to implement programmed load-disconnection logic without relying exclusively on mechanical relays.

Audible and Visual Alerts

The OLED display, LEDs, and buzzer provide different forms of status information. The display presents measured parameters and fault messages, while the LEDs provide immediate visual indications and the buzzer alerts the user to abnormal conditions. These features improve the usability of the system during operation and testing.

Battery Backup Support

The proposed system supports operation from the battery when the external DC adapter becomes unavailable. This feature is useful for backup applications in which short power interruptions should not immediately stop the connected load. The reliability of the backup function depends on the battery condition, load demand, and correct design of the power-transfer circuit.

Reduced Manual Supervision

The controller performs repeated measurements and evaluates the operating

condition automatically. The user does not need to check every parameter manually throughout operation. Although periodic maintenance and inspection remain necessary, automated monitoring can simplify the observation of battery condition.

Educational and Practical Value

The project combines embedded programming, sensor interfacing, power electronics, charging control, and fault detection. It helps demonstrate the relationship between measurement, control logic, and power switching in a practical engineering application. It also provides a foundation for future work involving data logging, wireless monitoring, and battery health estimation.

Adaptability to Industrial Applications

The design concept can be adapted for small UPS units, telecom backup systems, industrial control panels, emergency lighting, and low-voltage energy storage applications. Component ratings, charging characteristics, and protection settings must be changed as necessary for each application. The prototype should not be considered an industrial-certified protection system without appropriate design verification and safety testing.

RESULTS AND DISCUSSION

4.1 PROTOTYPE

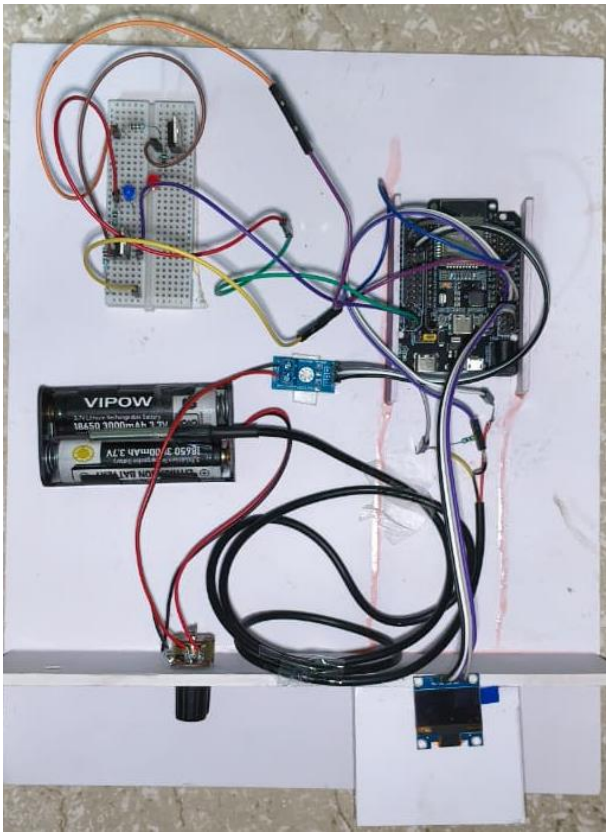


Fig 4.1 Prototype

The proposed Iot-Based Smart Battery Monitoring And Charging System is developed as a hardware prototype to demonstrate battery monitoring and protection functions. The prototype consists of an ESP32 DevKit V1, lithium-ion battery, INA219 current sensor, 0.96-inch OLED display, IRLZ44N power MOSFET, AMS1117-5.0 voltage regulator, potentiometer, blue LED, red LED, and breadboard. These components are interconnected to form a compact battery monitoring and protection system.

The ESP32 DevKit V1 acts as the main controller of the prototype. The lithium-ion battery provides the main power source, while the AMS1117-5.0 voltage regulator provides a regulated 5V supply. The INA219 sensor is used to monitor the current, voltage, and power parameters of the battery system. A potentiometer is connected to

provide an adjustable analog voltage input for simulating different battery voltage conditions during testing and demonstration.

The measured parameters are processed by the ESP32 and displayed on the 0.96-inch OLED display, allowing the user to observe the battery readings and system operating status in real time. The IRLZ44N power MOSFET acts as an electronic switching device to control the load according to the programmed protection conditions. Two LEDs are used to indicate the operating status of the system. The blue LED indicates a low-voltage condition, while the red LED indicates a fault condition.

The prototype is assembled using a breadboard, which provides a convenient platform for connecting the electronic components. During testing, the potentiometer is adjusted to simulate different voltage levels, and the ESP32 processes the input values to identify the corresponding operating conditions. The OLED display provides the monitoring information, while the LEDs indicate abnormal conditions. The power MOSFET demonstrates electronic load switching based on the programmed control logic.

Thus, the developed prototype demonstrates the basic functions of battery parameter monitoring, voltage-condition simulation, status indication, and MOSFET-based load control. It provides a practical demonstration of an ESP32-based intelligent battery monitoring and protection system.

Overall, the developed prototype successfully demonstrates the basic operation of an intelligent battery monitoring and protection system using ESP32. By integrating the INA219 sensor, potentiometer, OLED display, and power MOSFET, the system can monitor electrical parameters, simulate different battery voltage conditions, display real-time information, and control the load during programmed protection conditions. The blue and red LEDs provide clear visual indications of low-voltage and fault conditions.

Thus, the prototype demonstrates a simple and effective approach to battery monitoring and MOSFET-based protection for backup power and industrial applications.

4.2 SIMULATION OUTPUT

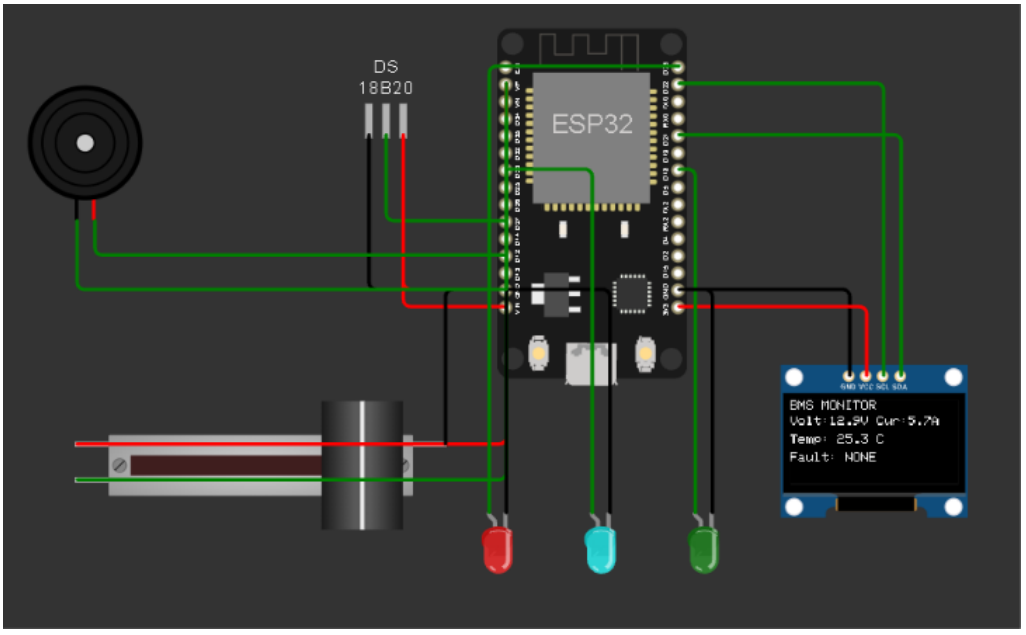


Fig 4.2 Simulation Output 1

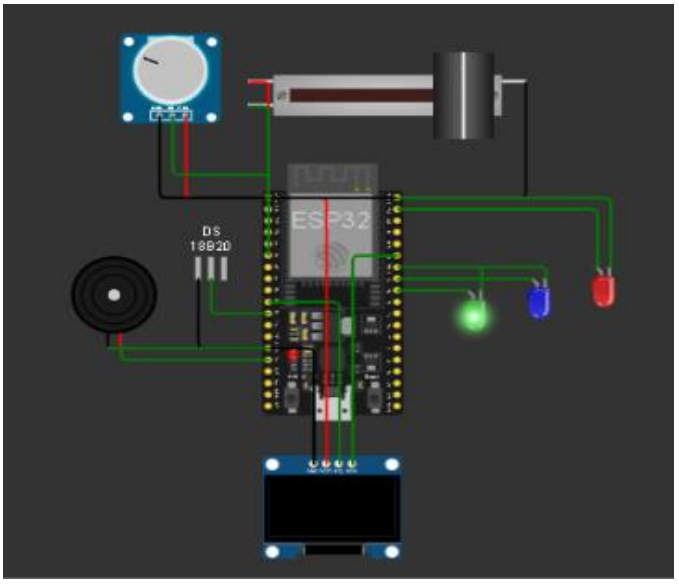


Fig 4.3 Simulation Output 2

The Wokwi simulation was used to verify the response of the system under different voltage, current, and temperature conditions. The ESP32 continuously reads the simulated sensor values and compares them with the programmed limits. The output condition is displayed on the OLED screen, while the LEDs and buzzer provide

additional indications. The system identifies the battery condition as normal, under-voltage, over-voltage, over-temperature, or over-current.

Normal Voltage Condition

When the battery voltage is between 9 V and 14.5 V, the system considers the voltage to be within the normal operating range. In this condition, the OLED displays the measured voltage and shows “FAULT: NONE.” The normal indicator LED remains active, and the buzzer remains switched off. The MOSFET control signal allows the load to remain connected. This result confirms that the system does not produce a false warning when the battery voltage is within the selected safe range.

Under-Voltage Condition

When the battery voltage becomes less than 9 V, the ESP32 identifies the condition as under-voltage. The OLED displays the measured voltage and the message “UNDER VOLTAGE.” The corresponding warning or fault LED is activated, and the buzzer provides an alert to the user. The MOSFET control can be changed to disconnect the load and prevent further battery discharge. This condition protects the battery from excessive discharge and indicates that the battery requires charging or inspection.

Over-Voltage Condition

When the battery voltage becomes higher than 14.5 V, the system identifies an over-voltage condition. The OLED displays the measured voltage and shows “OVER VOLTAGE.” The over-voltage indicator is activated, and the buzzer alerts the user. The MOSFET can disconnect the load or charging path according to the programmed control logic. This protection condition helps prevent damage caused by excessive voltage or an unsuitable charging condition.

Normal Temperature Condition

When the temperature remains at or below 37°C, the temperature is considered to be within the selected safe range. The OLED continues to display the measured temperature without showing an over-temperature warning. The indicator LEDs and

buzzer operate according to the voltage and current conditions. If the voltage and current are also within their limits, the OLED displays “FAULT: NONE,” and the load remains connected.

Over-Temperature Condition

When the temperature rises above 37°C, the ESP32 identifies the condition as over-temperature. The OLED displays the measured temperature and the message “OVER TEMP.” The warning or fault LED is activated, and the buzzer produces an alert. The MOSFET control may disconnect the load to reduce battery stress and prevent further temperature rise. This condition shows that the system can monitor temperature and provide a protective response when the selected temperature limit is exceeded.

Normal Current Condition

When the measured current remains below the programmed high-current limit, the current is considered normal. The OLED displays the current value, and no over-current warning is shown. The buzzer remains off unless another fault condition is present. The MOSFET remains enabled when the voltage and temperature values are also within their normal ranges.

Over-Current Condition

When the current exceeds the programmed high-current limit, the ESP32 identifies an over-current condition. The OLED displays the measured current and the message “OVER CURRENT.” The fault LED is activated, and the buzzer produces an audible warning. The MOSFET control signal can disconnect the load to protect the battery, wiring, and connected components from excessive current. This result confirms that the system can respond to an overload or possible short-circuit condition.

Combined Fault Conditions

If more than one abnormal condition occurs at the same time, the ESP32 checks all the measured values and displays the programmed fault message. For

example, if the voltage is below 9 V and the temperature is above 37°C, the system may display the under-voltage condition or the highest-priority fault according to the program logic. Similarly, if over-voltage and over-current occur together, the OLED can indicate the selected fault condition while the buzzer and LEDs provide warning signals.

The combined-condition response depends on the priority assigned in the program. A serious condition such as over-current may be given higher priority because it can cause rapid heating or damage. The system can also be programmed to disconnect the load whenever any protection limit is crossed.

Hardware Indication and Buzzer Working

The simulated hardware arrangement includes the ESP32, OLED display, DS18B20 temperature sensor, voltage control input, buzzer, and three status LEDs. These components work together to indicate the battery operating condition. The ESP32 processes the voltage, current, and temperature values and controls the LEDs and buzzer according to the detected condition..

The blue LED indicates standby mode. It glows when the system is powered and ready, but the battery is not actively charging and no fault condition is present. In this mode, the OLED continues to display the voltage, current, temperature, and system status. The buzzer remains off during normal standby operation.

The red LED indicates a fault condition. It glows when the ESP32 detects under-voltage, over-voltage, over-temperature, or over-current. During this condition, the OLED displays the corresponding fault message, and the MOSFET control can disconnect the load to protect the battery and connected components.

Thus, the green, blue, and red LEDs represent charging, standby, and fault conditions respectively. The buzzer provides additional confirmation for charging and an urgent warning during faults. The simulation confirms that the ESP32 can coordinate the LED, OLED display, and MOSFET control according to the programmed battery condition.

Discussion of Simulation Results

The simulation results show that the ESP32 can monitor the battery-related values and classify the operating condition according to the selected limits. Voltage values below 9 V are identified as under-voltage, values from 9 V to 12 V are treated as normal, and values above 14.5 V are identified as over-voltage. Temperatures above 37°C are identified as over-temperature, while current values above the programmed limit are identified as over-current.

The OLED display provides the measured values and the corresponding fault status. The LEDs and buzzer give additional visual and audible indications. The MOSFET control output represents the action taken to disconnect the load during an unsafe condition.

The results are obtained only from Wokwi simulation. They verify the programmed logic and the operation of the virtual display and indicators. They do not represent the performance of a physical battery, sensor, MOSFET, wiring system, or industrial load. A physical prototype would require further circuit design, sensor calibration, thermal testing, electrical testing, and safety validation.

BUDGET

Table 5.1 Budget

COMPONETS	BUDGET
Esp 32	450
Lithium Ion Battery	700
Temperature Sensor	120
Oled Display	250
Blue Led	10
Red Led	10
Power Mosfet	70
Voltage Regulator	40
Voltage Sensor	60
Bread Board	120
Resistor	40
Structure Board	150
Other	200
Total	2220

CONCLUSION AND FUTURE SCOPE

CONCLUSION

The Iot-Based Smart Battery Monitoring And Charging System was developed to monitor battery voltage, current, and temperature. The ESP32 acts as the main controller and compares the measured values with the programmed safety limits. Based on the detected conditions, the system displays battery status on the OLED and controls the MOSFET, LEDs, and buzzer. It identifies under-voltage below 9 V, normal voltage between 9 V and 14.5 V, over-voltage above 14.5 V, over-temperature above 37°C, and over-current conditions.

The system uses green, blue, and red LEDs to indicate charging, standby, and fault conditions, respectively. The buzzer provides audio alerts during charging and fault conditions. The Wokwi simulation demonstrates the ESP32 control logic and the operation of the display, indicators, buzzer, and MOSFET switching. Overall, the project provides a simple and expandable approach to battery monitoring and protection, forming a foundation for future physical implementation and further development.

The developed project is also aligned with Sustainable Development Goals SDG 7 (Affordable and Clean Energy) and SDG 9 (Industry, Innovation and Infrastructure). The prototype supports SDG 7 by demonstrating battery monitoring and controlled energy usage through voltage, current, and power measurement, along with MOSFET-based load protection. It supports SDG 9 by integrating ESP32, the INA219 sensor, and power MOSFET technology to develop an intelligent battery monitoring and protection system. The successful demonstration of these functions provides a foundation for further development of reliable battery backup and industrial energy storage systems.

FUTURE SCOPE

- The simulation can be converted into a physical prototype using actual voltage, current, and temperature sensors. The system can be tested with a real battery and load under controlled conditions.
- The switching circuit can be enhanced with a suitable gate driver, heat sink, fuse, and reverse-polarity protection. Thermal and fault testing can improve system safety and reliability.
- Wireless communication and data logging can be added using the ESP32. Battery parameters and fault information can be monitored remotely through a mobile application or computer.
- Future development can include battery state-of-charge estimation, cell balancing, and battery health monitoring. These features can improve battery performance, safety, and industrial applications.

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